END306 PROJECT DESCRIPTION

NS306 Productions Company

NS306 Productions is a small manufacturer that produces electronic parts used by a variety of other manufacturers. Recently, we noticed problems in a department that produces three different parts. The department is almost fully automated and consists of four lines. Each of the first three lines produces a single product and they have all been modified over time to meet the current demand mix. When the almost-finished parts (Parts A100, A200, and A300) exit these three lines, they all enter the fourth line where the final operations are performed for all product types. The new product mix has caused the fourth line to become a bottleneck.

NS306 Productions employed several consultants to collect data and recommend modifications to this line. Although we've received sufficient information to proceed with the modifications, the actual performance after the modifications are implemented is in question. Two different approaches were suggested. At this time, it's not clear which approach would provide the best results. It's also possible that both suggested modifications may be required to achieve the desired performance. Considering this uncertainty, we are initiating this request for recommendations. You are asked to evaluate the current design and propose the configuration(s) that will result in approximately minimizing operating cost while still achieving the desired performance levels. This document describes the current system, as well as including data where they are available. Your analysis should concentrate on the fourth line, and a schematic of it is shown in ***Figure 1***. We do not anticipate a need to change the first three lines, so we will give you only limited information on them.

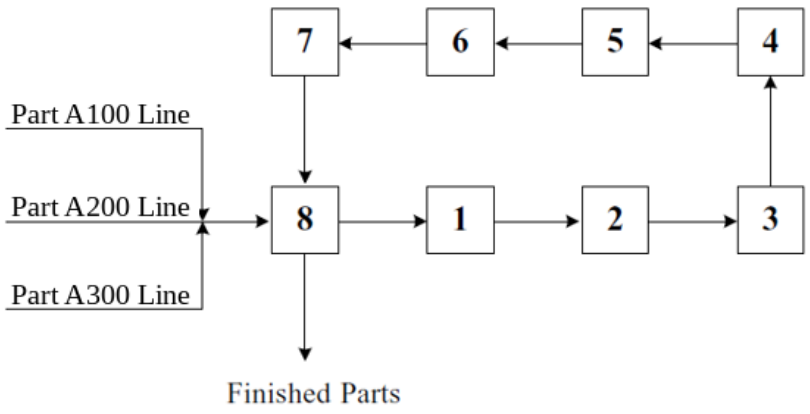


Figure 1. *System Schema*

Parts A100, A200, and A300 enter the fourth line on the left. Since the lines that produce these parts are fully automated, their arrival rates are very predictable. The normal arrivals for the three parts are: Part A100, every 2.8 minutes; Part A200, every 1.4 minutes; and Part A300, every 2.0 minutes. There are occasional jams on these lines, which can cause slight delays in part arrivals. The collected data show that delays occur 2%, 1.75%, and 0.5% of the time on Lines A100, A200, and A300, respectively. The data for these delays have been analyzed. The delays on the Part A100 line never fell below 10 seconds and never exceeded 30 seconds. Within this range, no value appeared to occur more often than any other; the observations were spread evenly between these two limits. The delays on the Part A200 and Part A300 lines behave in a similar manner, ranging between 15 and 35 seconds and between 15 and 40 seconds, respectively.

The arriving parts are merged (in the order in which they arrive) onto a single input conveyor, which feeds the fourth and final system. The input conveyor from the merge point to where the parts enter the fourth system has room for 40 parts. If this input conveyor becomes full and a new part attempts to merge onto the conveyor, then the line producing that part temporarily shuts down until space becomes available. Currently, there are numerous shutdowns during each day. For analysis purposes, you can assume that if a part arrives and the input conveyor is full, that part is considered to be lost production. The cost for a lost part is $0.96, $0.71, and $0.83 for Parts A100, A200, and A300, respectively.

The fourth system consists of eight work cells connected by a power-and-free conveyor. The distance between each work cell is 18 feet. When a new part enters the system from the input conveyor at Work Cell 8, it must wait for a special pallet that will hold the part as it travels through the system. Presently, there are 40 pallets in the system, each requiring 2 feet of space. Thus, there is room for eight pallets between each pair of successive work cells (16 feet of buffer space and 2 feet for the work cell).

When both a pallet and a new part are available at Work Cell 8, the finished part is first removed from the pallet, and then the new part is loaded. This unload/load process is fully automated and takes 25 seconds. A jam occurs approximately 5% of the time, requiring additional time. The time to respond to a jam never fell below 5 seconds and never exceeded 75 seconds. Response times of around 15 seconds were observed most often, and values became steadily less frequent the farther they were from 15 seconds in either direction. These parameters are expressed in seconds. Once the unload/load process is complete, the pallet moves to the buffer area in front of Work Cell 1. If there is no room in the buffer area, the pallet remains at the

unload/load work cell until room becomes available. In effect, the waiting loaded pallet blocks the unload/load work cell.

The new part moves through the system where an operation is performed at each of the remaining seven work cells. Work Cells 1, 3, 4, 5, and 6 are automated operations. Work Cells 2 and 7 are manual operations.

When a pallet enters an automated work cell, a scanner reads the part type on that pallet. If the part type is different from the last part type at that work cell, the work cell must undergo a setup. The required setup time is dependent only on the new part type. These setup times (in seconds) are given in ***Table 1***.

Arriving Part Type	Work Cell 1	Work Cell 3	Work Cell 4	Work Cell 5	Work Cell 6
Part A100	25	52	35	29	11
Part A200	20	21	22	14	19
Part A300	17	34	24	37	17

Table 1. *Setup Times*

For example, if a Part A200 arrives at Work Cell 1 and the previous part at that work cell was either an A100 or A300, then a 20-second setup is required. If the previous part was a A200, then no setup is required.

The operation or process time at the automated work cells is also part dependent. These operation times (in seconds) are given in ***Table 2***. For example, a Part A200 at Work Cell 4 requires a 38-second operation time.

Arriving Part Type	Work Cell 1	Work Cell 3	Work Cell 4	Work Cell 5	Work Cell 6
Part A100	37	39	41	33	31
Part A200	46	27	38	41	24
Part A300	39	23	47	35	51

Table 2. *Operation Times for automated work cells*

The operation or process times at the two manual work cells follow triangular distributions with the parameters given, in seconds. No setup is required at the manual work cells. The parameters are given in ***Table 3***.

Part Type	Work Cell 2	Work Cell 7
Part A100	36, 45, 52	27, 35, 41
Part A200	21, 32, 39	31, 39, 43
Part A300	32, 36, 42	22, 27, 28

Table 3. *Operation Times for manual work cells*

Based on our observations, we feel that you can safely assume that no jams or failures occur at Work Cells 1 through 7.

The factory is currently working two shifts, a 16-hour production day, five days a week. The lines are shut down at the end of each day, leaving the incomplete parts in the system, and they are restarted at the beginning of the next working day.

Although casual observations by the consultants and our production engineers indicate that our lost production cost is high, we are not able to place a dollar value on that loss. Thus, the first question we would like you to address is: What is the cost of production lost for our current system? Please provide this as a weekly cost.

NS306 Productions Company requests that during your analysis you evaluate and answer the following questions:

1. What is the cost of lost production for our current system?

2. What is the cost savings, if any, resulting from the implementation of alternative scenarios?

3. Should we implement any alternative scenarios?

Please provide all cost comparisons in dollars per week. Your report should include a recommendation for the most cost-effective system. Your analysis may reveal that none of the alternative options should be considered.